

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)Department of Artificial Intelligence and Data Science **ASSESSMENT****TOOL 1 – Data Interpretation**

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE-AI	Date:	

Code of Conduct

/ P. KALYANI (192472199) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **P. Kalyani**

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3

Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4
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Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots

1.1 Analysis of action-object relationships

Semantic representation converts a natural-language query into a structured meaning representation.

Customer Query	Semantic Representation	Action	Object	Meaning
Activate international roaming for my number.	ACTIVATE(Roaming, Customer)	ACTIVATE	Roaming	Customer wants roaming activated
Deactivate caller tune service.	DEACTIVATE(CallerTune, Customer)	DEACTIVATE	Caller Tune	Customer wants caller tune deactivated
Check my data balance.	QUERY(DataBalance, Customer)	QUERY	Data Balance	Customer wants to know remaining data
Enable 5G service.	ACTIVATE(5GService, Customer)	ACTIVATE	5G Service	Customer wants 5G activated

The representations follow an action-object relationship:

Action → Service/Object → Customer

For example:

ACTIVATE(Roaming, Customer)

means that the customer is requesting the activation of the roaming service.

1.2 Query containing semantic interpretation errors

From the chatbot logs:

Query ID	Actual Intent	Predicted Intent	Result
Q1	Activate Roaming	Activate Roaming	Correct
Q2	Deactivate Caller Tune	Activate Caller Tune	Incorrect
Q3	Check Data Balance	Query Data Balance	Correct
Q4	Enable 5G Service	Activate 5G Service	Correct

Q2 contains the semantic interpretation error.

The user says:

“Deactivate caller tune service.”

The correct representation should be:

DEACTIVATE(CallerTune, Customer)

but the system predicts:

ACTIVATE(CallerTune, Customer)

The object is correctly identified, but the action is reversed.

This is a serious error because activation and deactivation are opposite operations.

1.3 Effect of meaning representation on chatbot decision accuracy

Semantic representation directly affects the chatbot's decision-making.

For Q1:

ACTIVATE(Roaming, Customer)

- System identifies activation request
- Executes roaming activation
- Correct response.

For Q2:

ACTIVATE(CallerTune, Customer)

instead of:

DEACTIVATE(CallerTune, Customer)

- System may activate a service the customer wanted to cancel
- Incorrect backend action
- Poor customer experience.

From the four queries:

- Correct predictions = 3
- Incorrect predictions = 1

Therefore:

Accuracy = $(3/4) \times 100 = 75\%$

Error rate = $(1/4) \times 100 = 25\%$

Thus, even one semantic error can have significant consequences when the chatbot performs real service operations.

1.4 Recommended improvements

The telecom company should implement:

1. Intent classification

- Separate actions such as activate, deactivate, query, renew and cancel.

2. Context-aware semantic analysis

- Use previous conversation information to understand user intent.

3. Negation detection

- Explicitly identify words such as *not*, *deactivate*, *cancel*, *stop*, and *disable*.

4. Domain-specific ontology

- Maintain relationships between telecom services and actions.

5. Entity recognition

- Identify entities such as roaming, 5G, caller tune, data balance and phone number.

6. Confidence scoring

- If confidence is low, ask the customer for confirmation.

7. Confirmation for critical actions

- For example:

“Do you want me to deactivate your caller tune service?”

Conclusion

Semantic representation provides the bridge between natural language and chatbot actions. The data shows 75% prediction accuracy, but Q2 demonstrates that confusing the action can lead to an incorrect service operation. A combination of intent classification, contextual understanding, negation detection and confirmation mechanisms would improve reliability.

2. First-Order Predicate Calculus for Smart Manufacturing

2.1 Representing production data using First-Order Predicate Calculus

The production data is:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

We can represent the facts using predicates:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

The rules are:

Rule 1

Active machine $\rightarrow$ Producing

$\forall x [\text{Active}(x) \rightarrow \text{Producing}(x)]$

Therefore:

Producing(M1)

Producing(M2)

Producing(M4)

Rule 2

The given rule:

$$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$$

can be written as:

$$\forall x \forall y [(\text{Produces}(x,y) \wedge \text{Active}(x)) \rightarrow \text{Available}(y)]$$

This means that if an active machine produces a product, that product is available.

Rule 3

$$\forall x [\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)]$$

Therefore:

$$\neg \text{Producing}(M3)$$

2.2 Which products are currently available?

The question requires the $\text{Produces}(x,y)$ facts to determine exact products.

However, no machine-product mapping is provided in the production data.

Therefore, we cannot logically conclude that a specific product such as Gear is available unless a fact such as:

$$\text{Produces}(M1, \text{Gear})$$

is provided.

For example, if:

$$\text{Produces}(M1, \text{Gear})$$

and:

$$\text{Active}(M1)$$

then:

Available(Gear)

can be inferred.

Therefore, the correct logical conclusion from the supplied data is:

Specific available products cannot be determined because the Produces(x,y) relationships are missing.

This is an important limitation in the dataset.

2.3 Is Gear production affected by maintenance?

Again, the available information states:

Maintenance(M3)

Therefore:

$\neg$ Producing(M3)

If M3 produces Gear, i.e.:

Produces(M3,Gear)

then Gear production by M3 is affected.

However, because the dataset does not explicitly state:

Produces(M3,Gear)

we cannot conclude that overall Gear production is affected.

Logical conclusion

Conditional conclusion:

If M3 is responsible for producing Gear, its maintenance activity stops that machine from producing Gear.

But based strictly on the supplied data, Gear production impact cannot be definitively established.

This demonstrates the importance of complete relational data in predicate-based decision systems.

2.4 Effectiveness of predicate logic in manufacturing

First-Order Predicate Calculus is useful because it provides:

1. Precise representation

Machine states can be represented formally.

2. Rule-based inference

The system can automatically derive:

$\text{Active}(\text{M1}) \rightarrow \text{Producing}(\text{M1})$

3. Maintenance monitoring

$\text{Maintenance}(\text{M3}) \rightarrow \neg \text{Producing}(\text{M3})$

allows the system to identify machines that are unavailable for production.

4. Explainability

A decision can be traced through logical rules.

5. Automation

The same rules can be applied continuously to production data.

Limitations

Predicate logic alone does not provide:

- probabilistic reasoning
- uncertainty handling
- sensor noise management
- real-time prediction
- machine-learning-based forecasting

Therefore, industrial systems can combine predicate logic with IoT sensors, databases, machine learning and rule engines.

Conclusion

Predicate calculus provides a clear and explainable framework for manufacturing decision support. However, the current dataset lacks the machine-to-product Produces relationships, so exact product availability and Gear-production impact cannot be determined without additional facts.

3. Word Sense Disambiguation in E-Commerce Search Engines

3.1 Determine the correct word sense

The correct senses can be identified using the search context and clicked results.

Query	Possible Senses	Clicked Result	Correct Sense
Apple accessories	Fruit / Technology Brand	iPhone Charger	Technology Brand
Mouse wireless	Animal / Computer Device	Bluetooth Mouse	Computer Device
Java tutorial	Island / Programming Language	Coding Lessons	Programming Language
Python course	Snake / Programming Language	Software Development Training	Programming Language

Thus, all four queries demonstrate technology-related meanings.

3.2 Semantic cues used for disambiguation

Apple accessories

The word “accessories” is a strong commercial cue.

The clicked result:

iPhone Charger

confirms that Apple means the technology company.

Mouse wireless

The word “wireless” is associated with computer hardware.

A Bluetooth Mouse is a technological device, not an animal.

Java tutorial

The word “tutorial” is a strong educational cue.

“Coding Lessons” indicates that Java refers to the programming language.

Python course

The word “course” combined with “software development training” indicates the programming language.

Therefore, the system uses:

- neighboring words
- query intent
- product category
- clicked results
- historical user behavior
- domain context

for disambiguation.

3.3 Impact of incorrect sense selection

Incorrect WSD can negatively affect the search engine.

Example 1

Query:

“Apple accessories”

If the system selects fruit, it might return:

- fruit baskets
- apple recipes
- agricultural products

instead of:

- iPhone chargers
- AirPods accessories
- Apple Watch accessories

Example 2

Query:

“Python course”

If interpreted as a snake, the system may return:

- snake-care courses
- reptile information
- animal-related products

instead of programming courses.

This results in:

Wrong sense → Wrong results → Lower relevance → Fewer clicks → Lower conversion rate → Poor user satisfaction

3.4 Industrial-scale WSD strategy

An e-commerce company should use a hybrid contextual WSD system.

Proposed architecture

User Query



Tokenization



Context Analysis



Named Entity Recognition



Candidate Sense Generation



Contextual Embedding Model



User/Search History



Sense Ranking



Product Retrieval



Recommendation Ranking

Techniques

1. Contextual embeddings

Transformer-based models can understand how word meaning changes based on surrounding words.

2. Product ontology

Maintain categories such as:

Apple

└─ Technology Brand

└─ Fruit

Mouse

└─ Computer Device

└─ Animal

3. Search history

Previous user behavior can help identify intent.

4. Click-through data

Clicked products provide useful feedback.

5. Feedback learning

The system can learn from:

- clicks
- purchases
- search reformulation

- abandoned results
- product views

Conclusion

The four queries show that context is essential for word sense disambiguation. Correct WSD improves search relevance, recommendations, customer satisfaction and potentially conversion rates.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

4.1 How syntactic structures contribute to semantic role assignment

The sentences are:

1. Doctor prescribed medicine to patient.
2. Patient reported severe headache.
3. Nurse monitored patient continuously.
4. Medicine reduced blood pressure.

Syntactic structure identifies relationships between words.

For example:

Doctor prescribed medicine to patient.

Structure:

Subject → Verb → Object → Recipient

Therefore:

- Doctor → Agent
- Medicine → Theme
- Patient → Recipient

Similarly:

Patient reported severe headache.

- Patient → Experiencer/Agent
- Headache → Reported symptom

Thus, syntax provides the structural foundation for semantic-role labeling.

4.2 Are the assigned semantic roles appropriate?

The provided roles are:

Entity	Assigned Role	Evaluation
Doctor	Agent	Appropriate
Medicine	Instrument	Not appropriate
Patient	Recipient	Appropriate for first sentence
Headache	Symptom	Appropriate

Important correction: Medicine

In:

“Doctor prescribed medicine to patient.”

Medicine is better represented as the Theme/Object, not an Instrument.

An instrument is something used to perform an action, such as:

“The doctor used a syringe.”

Here:

- Doctor → Agent

- Syringe → Instrument
- Medicine → Theme
- Patient → Recipient

Therefore, assigning Medicine = Instrument is a semantic-role error.

4.3 Potential errors from incorrect parsing

Incorrect syntactic parsing can lead to incorrect clinical interpretation.

Example

Correct:

Doctor → prescribed → Medicine

Doctor → prescribed → Medicine → to Patient

If the parser incorrectly identifies patient as the subject, the system could interpret:

Patient prescribed medicine to doctor.

This completely changes the clinical meaning.

Possible consequences

- incorrect medical records
- wrong treatment extraction
- incorrect patient history
- wrong clinical decision support
- incorrect alerts
- reduced doctor trust
- potential patient-safety risks

Therefore, syntax errors in healthcare NLP can have much more serious consequences than ordinary search errors.

4.4 Recommended methods for improvement

1. Dependency parsing

Use dependency parsing to identify relationships between words.

For example:

Doctor
↓ nsubj
prescribed
↓ dobj
medicine
↓ nmod
patient

This provides detailed grammatical relationships.

2. Semantic Role Labeling

Use SRL to identify:

- Agent
- Patient/Theme
- Recipient
- Experiencer
- Instrument
- Location

- Time

3. Medical Named Entity Recognition

Recognize entities such as:

- Doctor
- Patient
- Medicine
- Disease
- Symptom
- Treatment
- Hospital
- Location

4. Medical ontology

Use domain knowledge to distinguish:

Medicine $\neq$ Instrument

Patient $\neq$ Treatment

Disease $\neq$ Medication

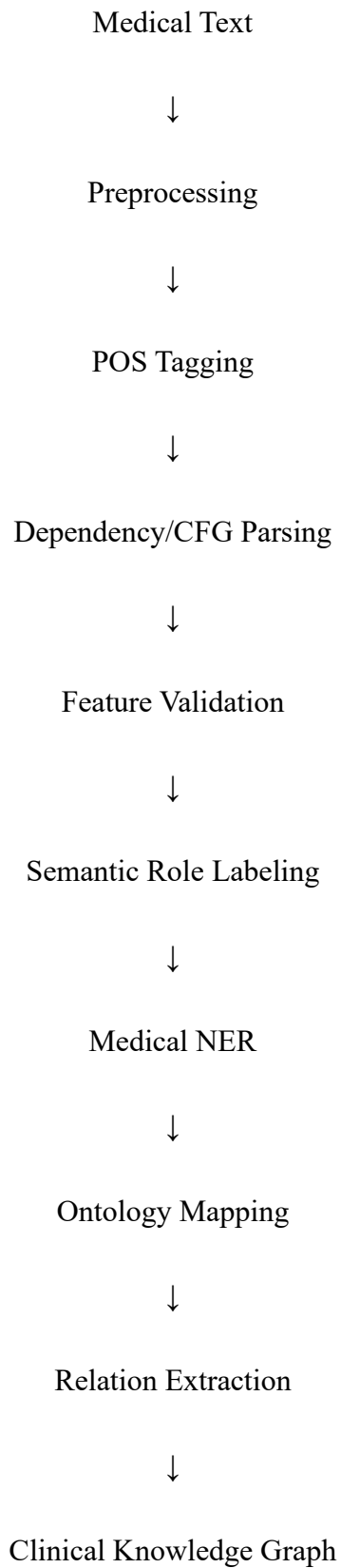
5. Domain-specific parsing

Generic parsers may not understand specialized medical terminology.

A healthcare system should use medical corpora and domain-specific models.

6. Hybrid syntax-semantic architecture

A strong architecture is:



This reduces errors caused by relying only on syntax.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____